

## Ch. 3 — P02 (C) Model Boundary Decisions

Problem Bank | Chapter 3 | Type: C | Difficulty:

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For each scenario below, a simulation analyst has made a boundary decision. Evaluate whether the boundary is appropriate given the study question. If the boundary is wrong, explain what is missing and what error it introduces.

1. **Study question:** “What is the mean patient wait in the ER?”

**Boundary:** Patients arrive, wait for a bed, see a physician, and leave. Lab tests are treated as an instantaneous delay of fixed duration.

**Omitted:** Lab processing time variability.

Is this boundary appropriate? Under what conditions would omitting variability matter most?

2. **Study question:** “Does adding a second baggage carousel reduce passenger pickup wait?”

**Boundary:** Passengers arrive at the carousel area after their flight lands, wait for their bag, and leave.

**Omitted:** The flight landing process and the bag loading process.

Evaluate this boundary. Is the omitted process relevant?

3. **Study question:** “How does a change in reorder policy affect inventory costs?”

**Boundary:** Demand arrives, is filled from on-hand inventory, and backorders accumulate if stock-out occurs. Orders arrive after a lead time drawn from a distribution.

**Omitted:** The supplier’s production process.

Is the supplier’s process inside or outside the natural boundary for this question?

4. **Study question:** “How long does a patient spend in the hospital?”

**Boundary:** Patient is admitted, assigned a bed, receives daily care, and is discharged when clinically ready.

**Omitted:** Post-discharge home care and readmission within 30 days.

Classify: is readmission a boundary decision or a simplifying assumption? What would you need to know to decide whether to include it?